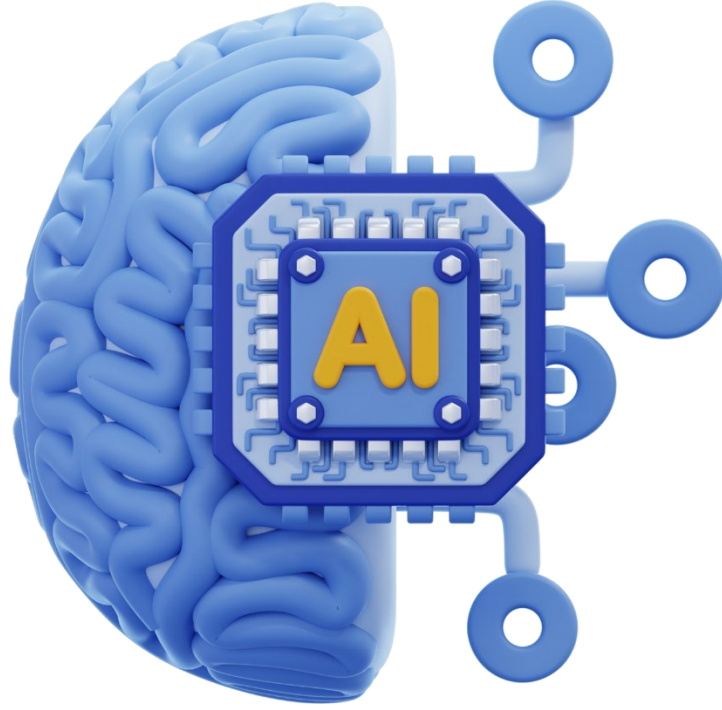
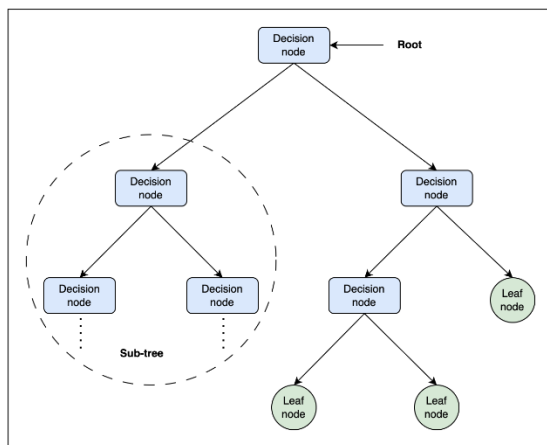




Decision Tree

اللَّهُمَّ صَلِّ عَلَى مُحَمَّدٍ وَعَلَى آلِ مُحَمَّدٍ، كَمَا صَلَّيْتَ عَلَى إِبْرَاهِيمَ، وَبَارِكْ عَلَى مُحَمَّدٍ وَعَلَى آلِ مُحَمَّدٍ،
كَمَا بَارَكْتَ عَلَى آلِ إِبْرَاهِيمَ، فِي الْعَالَمِينَ، إِنَّكَ حَمِيدٌ مَجِيدٌ

- ✚ **Decision forest** models are **composed of decision trees**. Decision forest learning algorithms (like random forests) rely, at least in part, on the learning of decision trees.
- ✚ **Decision tree** is a model composed of a collection of "questions" organized **hierarchically** in the shape of a **tree**. The questions are usually **called a condition**, a split, or a test.
- ✚ Each **non-leaf** node contains a condition, and each **leaf** node contains a **prediction**.
- ✚ Decision trees are usually represented with the **root** (the **first** node) at the top.
- ✚ **Inference** of a decision tree model is computed by routing an example from the root (at the top) to one of the leaf nodes (at the bottom) according to the conditions. The value of the reached leaf is the decision tree's prediction. The set of visited nodes is called the **inference path**.

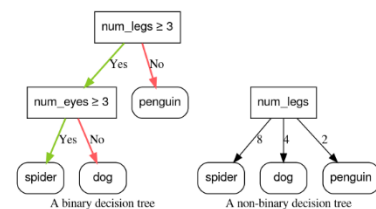


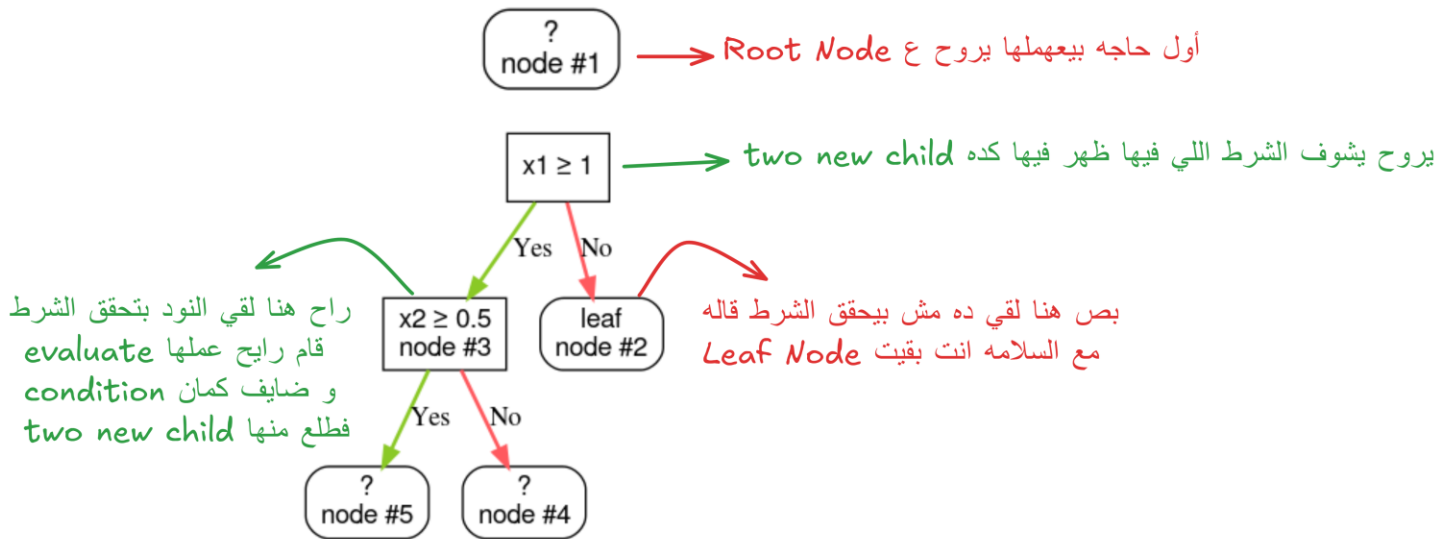
اول Condition في **Decision Tree** بيكون عبارته عن **Root Node** كل **condition** node بيكون له **Children** عبارته عن **prediction** آخر نود بنوصلها دي بتكون عبارته عن **leaf node** يكون نود مفيهاش أي **condition** و انما هي عبارته عن **prediction** تمام ال **Path** اللي باخده من **Root Node** لحد ال **leaf nodes** اللي هي ال **prediction** د عبارته عن **inference path**.

- ✚ Decision trees **can predict numerical** values by labeling leaves with regressive predictions (numerical values).
- ✚ **Axis-Aligned Condition**: This type of condition in decision trees **involves only one feature** at a time.
- ✚ **Oblique Condition**: This type of condition involves **multiple features combined in a single condition**.
- ✚ Decision trees with **binary decision trees**: Conditions with **two** possible outcomes.
- ✚ Decisions With **non-binary decision trees**: have **more than two possible** outcomes.

❖ Decision Tree Training:

- Decision trees are trained to best explain a set of training examples.
- Most algorithms used to train decision trees work with a greedy **divide and conquer strategy**.
- At each node, all the possible conditions are evaluated and scored. The algorithm selects the "**best**" condition, that is, the condition with the **highest score**.
- The algorithm then repeats recursively and independently on both children nodes. When no satisfying conditions are found, the node becomes a **leaf**.
- The routine responsible for finding the best condition is called the **splitter**. Because it needs to test a lot of possible conditions, splitters are the bottleneck when training a decision tree.





❖ Attribute Selection Measures:

- While implementing a Decision tree, the main issue arises that how to select the best attribute for the root node and for sub-nodes.
- To solve such problems there is a technique which is called as **Attribute selection measure** or **ASM**.
- **Information Gain:**
 - ✓ Measures the **reduction in entropy after a dataset split**.
 - ✓ Selection Criterion: The attribute with the **highest information gain is chosen** for the split.
 - ✓ Goal: **Maximize** information gain for better splits.
- **Gini Index:**
 - ✓ Measures the **probability of mislabeling** a random sample if it is labeled randomly based on the set's distribution.
 - ✓ Selection Criterion: The **change in the Gini index** before and after a split is used to choose the split attribute.
 - ✓ Goal: Select an attribute that **minimizes** the Gini index to improve purity in nodes.

$$Entropy(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$$

لو عندي ان **Predict** الشخص ده عنده **Cancer** او لا ف P_- هنبقي احتمال انه مش مصاب و P_+ احتمال انه مصاب و S هو عدد ال **Training Examples**

- ✓ **Zero Entropy:** Entropy is zero when **all examples are uniformly positive or negative** (pure set).
- ✓ **Maximum Entropy:** Entropy is one when there **are equal numbers of positive and negative examples** (maximum disorder).
- ✓ **Information Gain:** Measures the reduction in entropy when **data is split by an attribute**, guiding the selection of the best attribute for splitting.
- ✓ **Entropy:** measure of the impurity or uncertainty of a set of data. It ranges from 0 (completely pure) to 1 (completely impure).

- ✓ **Entropy in Information Theory:** Measures the impurity or unpredictability within a group of observations.

انا بسخدم ال **Entropy** ليه أصلا ؟ مش انا عندي **Feature** كتير في الداتا سيت بتاعتي هل كله بيأثروا او بيساعدوا الموديل انه يقدر يطلع النتيجة ال **Predicted** بطريقة صحيحة ؟ لا فانا محتاج اشوف طريقه تساعدني اشوف تأثير كل **Feature** بحيث اعرف انا هعتمد ع ايه بالظبط فهنا بقي يجي دور ال **Entropy** اللي بناء ع قية ال **Feature** اللي هتطلع لو بتقرب ل 0 يبقى دي **Feature** دي مهم لو بتقرب 1 تبقي بتأثر تأثير ضعيف .

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

هنا قالك انا هشوف أهمية عمود ال **Wind** بإستخدام **Gain**

Dataset					
Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

فهنا عمود ال **Wind** بيحتوي ع قيمتين اما **Weak, Strong**

$Values(Wind) = \{Weak, Strong\}$

$$S = [9+, 5-]$$

$$S_{Weak} = [6+, 2-]$$

$$S_{Strong} = [3+, 3-]$$

قيمة ال **Weak** قالك هاخذها ب + عندي كام واحد ب **Yes**

قالك 9 فحطها ب 9+ طب كام واحد ب **No** قالك 5 فحطها ب 5-

جه هنا تاني قسم قالك ال **Weak** في كام قيمة ليها ب **Yes** طلع ب 6

طب كام قيمة ليها ب **No** طلع ب 2- و عمل كده مع **Strong** شاف

كام قيمة ب **Yes** حطها موجب و كام قيمة **No** حطها سالب

$$Gain(S, Wind) = Entropy(S) - \sum_{v \in \{Weak, Strong\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Entropy(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$$

$$Entropy(S) = -\frac{9}{14} \log_2 \left(\frac{9}{14}\right) - \frac{5}{14} \log_2 \left(\frac{5}{14}\right) = 0.94$$

تعالى نجيب ال **Entropy** ف + اللى هي فوق كانت ل 9 هنقسمها

ع عدد القيم اللي عندنا في **Train** اللى هم 14 قيمة طلعت ب 9 / 14

اما **P-** اللى هي **Strong** ل 5 / 14

$$\frac{|S_{Weak}|}{S} = \frac{8}{14} \quad Entropy(S_{Weak}) = -\frac{6}{8} \log_2 \left(\frac{6}{8}\right) - \frac{2}{8} \log_2 \left(\frac{2}{8}\right) = 0.811$$

$$\frac{|S_{Strong}|}{S} = \frac{6}{14} \quad Entropy(S_{Strong}) = -\frac{3}{6} \log_2 \left(\frac{3}{6}\right) - \frac{3}{6} \log_2 \left(\frac{3}{6}\right) = 1$$

$$Gain(S, Wind) = 0.94 - \left(\frac{8}{14} Entropy(S_{Weak}) + \frac{6}{14} Entropy(S_{Strong})\right)$$

$$Gain(S, Wind) = 0.94 - \left(\frac{8}{14}(0.811) + \frac{6}{14}(1)\right) = 0.048$$

علشان نطبق قانون **Gain** هنجيب بقي **Entropy** ل **Weak** لوحده و ل

Strong لوحده فعندنا كام واحد **Weak** معاها **Yes** قالك 6 يبقى دي **P+** طب

كام واحد **Weak** ب **No** قالك 2 فدي ب **P-** طب هنقسم قيم **P** ع 8 ليه علشان احنا

شغالين علي **Weak** بس اللي كله ب 8 مش ع الداتا كلها

طب يا باشا قانون ال **Gain** طلع معايا ال **Gain** الخاص ب **Feature** ال **Wind** ب 0.048

$$Gain(S, Outlook) = 0.2464$$

$$Gain(S, Temp) = 0.0289$$

$$Gain(S, Humidity) = 0.1516$$

$$Gain(S, Wind) = 0.048$$

هنطبق بقي ال **Gain** بنفس الطريقه ع كل

Feature اللي عندي و هيطلع عندي القيم

دي هناخد أكبر قيمة من **Features** دي

انها تبقي **Root Node**

هنا قالك انا هشوف أهمية عمود ال Wind بإستخدام ال GINI

□ Compute The Gini index for the Wind attribute

$$Gini(Weak) = 1 - \sum_{i=1}^c (P_i)^2 = 1 - \left(\left(\frac{6}{8} \right)^2 + \left(\frac{2}{8} \right)^2 \right) = 0.375$$

قالك انا هشتغل الاول علي Weak فقالك ال Weak اصلا كلهك ع بعض 8 فيهم 6 ب Yes و 2 ب No فطلع ال Gini بتاعه ب 0.375

$$Gini(Strong) = 1 - \sum_{i=1}^c (P_i)^2 = 1 - \left(\left(\frac{3}{6} \right)^2 + \left(\frac{3}{6} \right)^2 \right) = 0.5$$

اما في Strong فهو في منه 6 عناصر منهم 3 ب Yes و 3 ب No فطلع ب 0.5

$$Gini(Wind) = W_1 Gini(Weak) + W_2 Gini(Strong) = \frac{8}{14} \times 0.375 + \frac{6}{14} \times 0.5 = 0.428571$$

لما جه يحسب ال Gini للعمود بالكامل هتلاقي ضرب Weight جابه من قيمة ال Weight دي قالك تعالي الاول بص عندك كام عنصر Weak في عمود ال weak قالك 8 عناصر من اصل 14 فقسم 8 ع 14 و ال strong من اصل 14 فطلع عندك قيمة ال Gini للعمود Wind ب 0.428

$$Gini(outlook) = 0.7143$$

$$Gini(Temp) = 0.4403$$

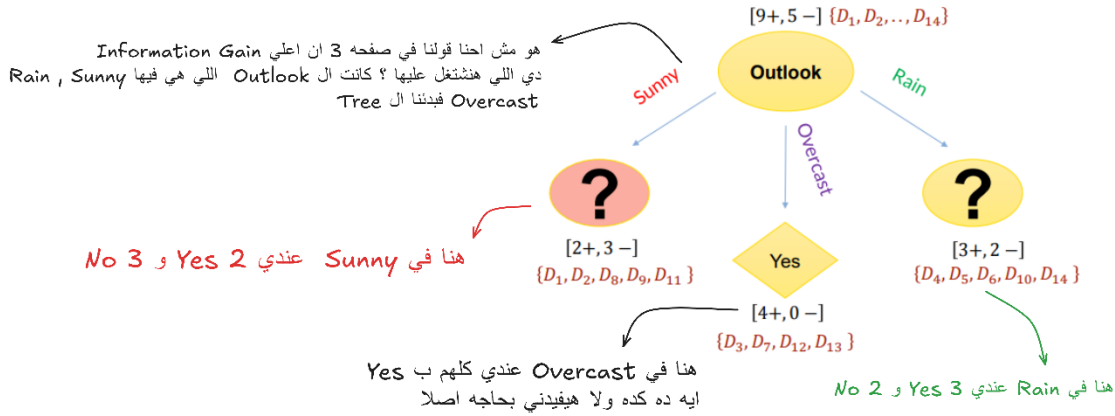
طبقي بقي ال GINI علي كل الصفوف و هناخد ال Feature اللي له أقل قيمة Gini

$$Gini(Humidity) = 0.36735$$

$$Gini(Wind) = 0.428571$$

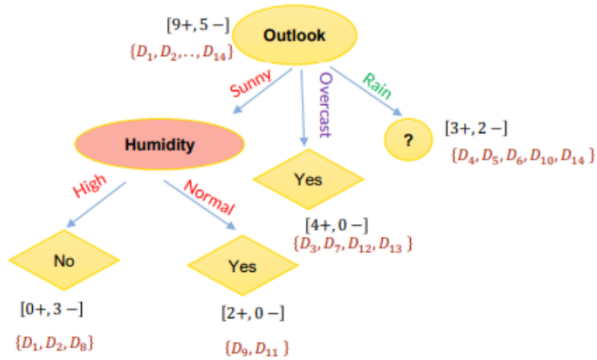
❖ Decision Tree Learning Algorithms:

- **ID3:** Iterative Dichotomies 3: This algorithm leverages entropy and information gain as metrics to evaluate candidate splits.
- **C4.5:** This algorithm is considered a later iteration of ID3
- **CART:** The term, CART, is an abbreviation for “classification and regression trees”. This algorithm typically utilizes Gini impurity to identify the ideal attribute to split on.



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D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

قالك خلاص شيل اي Row عبارة عن Overcast اصلا بدل كده مش هياثر بيشي كل ال Rows اللي فيها Overcast اصبح عندي دلوقتي اتجاهين اما اروح ل Sunny او اني اروح ل Rain



بالنسبه ل Sunny قالك انا هروح ارجع الداتا اللي فيها Sunny بس و احسبها Gain

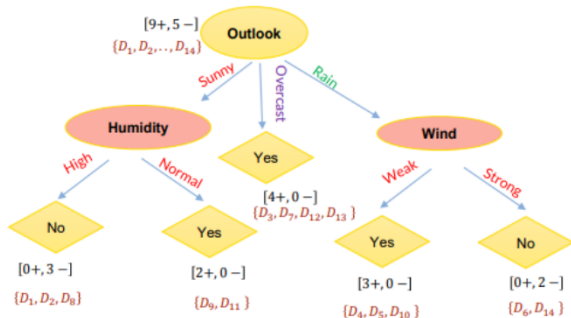
Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes

حسبنا ال Gain عادي خالص زي ما عملنا فوق طلع عندنا اعلي Gain ل Humidity فده اللي هروح له من عند ال Sunny

$$Gain(S, Wind) = 0.0192$$

$$Gain(S, Humidity) = 0.97$$

$$Gain(S, Temp) = 0.570$$



بالنسبه ل Rain قالك انا هروح ارجع الداتا اللي فيها Rain بس و احسبها Gain

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D10	Rain	Mild	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

حسبنا ال Gain عادي خالص زي ما عملنا فوق طلع عندنا اعلي Gain ل Wind فده اللي هروح له من عند ال Rain

$$Gain(S, Wind) = 0.97$$

$$Gain(S, Humidity) = 0.0192$$

$$Gain(S, Temp) = 0.0192$$